

Internship Report – Day 3

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Internship Program: ParkNSecure – Smart Parking Solutions

Day 3 Overview:

Today, I developed a Python script using OpenCV that accesses the webcam, detects license plates in real time, and automatically captures and saves the detected plate region as an image in a local folder. The script processes each video frame by converting it to grayscale, applying a bilateral filter, and using adaptive thresholding to enhance edge detection. Contours are identified and filtered based on shape and aspect ratio to locate potential license plates. A cooldown mechanism prevents duplicate captures by checking the time and location of the last detection. Detected plates are highlighted with colored rectangles (green for new, yellow for recent), and each capture is saved with a timestamped filename, forming a base for further enhancements like OCR integration.

Tools Used Today:

VS CODER, GEMINI, COLAB, SONNET.

Task Given:

TO CREATE THE ANPR MODEL TO DETECT THE LETTERS AND NUMBERS PRESENT IN THE VEHICLES NUMBERPLATES.

Approach to the Task:

I wrote a Python script using OpenCV that opens the webcam, detects the number plate in the video stream, and captures that specific region as a photo. Once the number plate is identified based on its shape and size, the script crops that area from the frame and saves it automatically to a folder. This helps in directly storing the number plate image without saving the entire frame.

Attachments:-

```
license_plate_detector.py > ...
1  import cv2
2  import numpy as np
3  from datetime import datetime
4  try:
5      import pytesseract
6      from PIL import Image
7      from pytesseract import TesseractNotFoundError
8  except ImportError:
9      print("Warning: pytesseract and/or Pillow (PIL) not installed. OCR functionality will be disabled.")
10 except TesseractNotFoundError:
11     print("Warning: Tesseract is not installed or not in your PATH. OCR functionality will be disabled.")
12
13 def recognize_license_plate():
14     """Captures a single frame, detects a potential license plate with more filtering, saves it, and attempts OCR."""
15     cap = cv2.VideoCapture(0)
16     cap.set(cv2.CAP_PROP_FRAME_WIDTH, 1280)
17     cap.set(cv2.CAP_PROP_FRAME_HEIGHT, 720)
18
19     potential_plate_filename = None
20
21     while True:
22         ret, frame = cap.read()
23         if not ret:
24             print("Error: Could not read frame.")
25             break
26
27         display_frame = frame.copy()
28         gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
29         filtered = cv2.bilateralFilter(gray, 11, 17, 17)
30         thresh = cv2.adaptiveThreshold(
31             filtered, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
32             cv2.THRESH_BINARY_INV, 11, 2
33         )
34         contours, _ = cv2.findContours(
35             thresh, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE
36         )
37         contours = sorted(contours, key=cv2.contourArea, reverse=True)[:50] # Check Top 50 largest
38
39         plate_found = False
40         for contour in contours:
41             peri = cv2.arcLength(contour, True)
42             approx = cv2.approxPolyDP(contour, 0.018 * peri, True)
43
44             if len(approx) == 4:
45                 x, y, w, h = cv2.boundingRect(contour)
46                 aspect_ratio = w / float(h)
47                 if 2.0 <= aspect_ratio <= 5.0:
48                     # More advanced filtering based on solidity and extent
49                     area = cv2.contourArea(contour)
50                     hull = cv2.convexHull(contour)
51                     hull_area = cv2.contourArea(hull)
52                     solidity = float(area) / hull_area if hull_area > 0 else 0
53
54                     rect_area = w * h
55                     extent = float(area) / rect_area if rect_area > 0 else 0
56
57                     # Adjust these thresholds based on your environment and plate characteristics
58                     if solidity > 0.3 and extent > 0.6:
59                         cv2.rectangle(display_frame, (x, y), (x + w, y + h), (0, 255, 0), 2)
60                         plate_img = frame[y:y + h, x:x + w]
61                         timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
62                         potential_plate_filename = f"potential_plate_{timestamp}.jpg"
63                         cv2.imwrite(potential_plate_filename, plate_img)
64                         print(f"Potential license plate saved as: {potential_plate_filename}")
65                         plate_found = True
66                         break # Exit after finding one potential plate that passes filters
67
68         cv2.imshow('Capture Frame (Press "s" to save and recognize, "q" to exit)', display_frame)
69
70         key = cv2.waitKey(1) & 0xFF
71         if key == ord('s') and plate_found:
72             break
73         elif key == ord('q'):
74             break
75
76     cap.release()
77     cv2.destroyAllWindows()
78
79     if potential_plate_filename:
80         try:
81             img = Image.open(potential_plate_filename)
82             text = pytesseract.image_to_string(img)
83             print("Recognized Number Plate:", text.strip())
```

```

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80     try:
81         img = Image.open(potential_plate_filename)
82         text = pytesseract.image_to_string(img)
83         print("Recognized Number Plate:", text.strip())
84     except ImportError:
85         print("Error: pytesseract or Pillow not installed. Cannot perform OCR.")
86     except TesseractNotFoundError:
87         print("Error: Tesseract not installed or not in your PATH. Cannot perform OCR.")
88     except FileNotFoundError:
89         print(f"Error: Could not find the saved plate image: {potential_plate_filename}")
90     except Exception as e:
91         print(f"An error occurred during OCR: {e}")
92
93 if __name__ == "__main__":
94     recognize_license_plate()

```



```
Potential license plate saved as: potential_plate_20250507_112726.jpg
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Traceback (most recent call last):
  File "C:\Users\C H UDAY KUMAR\OneDrive\Desktop\parkNsecure\ANPR\license_plate_detecto
n <module>
```